

ReconUNet: Learning to Reconstruct Clean Covariance matrix under Low SNR, Multipath and Array imperfections

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Abstract—Direction-of-Arrival (DOA) estimation is a fundamental task in array signal processing, it is known to degrade in practical regimes characterized by low signal-to-noise ratios, limited snapshots, source coherence, and array mismatches such as gain/phase errors, sensor position bias, and mutual coupling. We propose a data-driven front-end that restores a subspace-faithful covariance prior to applying classical estimators. The method ingests multi-lag autocorrelation features and uses a U-Net backbone with lightweight convolutional and Multilayer-Perceptron (MLP) heads to reconstruct a denoised, mismatch-compensated covariance matrix; its Eigen-Decomposition (EVD) is produced explicitly and supplied to downstream algorithms. Training combines a mean-squared reconstruction loss with a subspace-similarity objective that aligns the predicted and reference signal subspaces. Comprehensive experiments cover varying numbers of sources, a broad span of Signal to Noise Ratio (SNR), both coherent and non-coherent conditions, and realistic levels of array imperfections. Using root-mean-square angular error as the primary metric, we benchmark beamforming, Minimum Variance Distortionless Response (MVDR), multiple signal classification (MUSIC), Root-MUSIC, Estimation of signal parameters via rotational invariant techniques (ESPRIT) and Unitary-ESPRIT with and without the proposed front-end. A single trained model generalizes across operating conditions and consistently improves downstream accuracy relative to applying the classical methods directly to raw sample statistics. These results indicate that learning to reconstruct the covariance—rather than regressing angles end-to-end—provides a practical, plug-and-play enhancement to classical DoA pipelines. By restoring separation between signal and noise subspaces under coherence and model mismatch, the approach yields robust, accurate estimates while preserving compatibility with standard subspace-based techniques and producing an explicit EVD useful for subsequent processing.

Index Terms—Direction-of-Arrival (DOA), array signal processing, deep learning, subspace methods

I. INTRODUCTION

Direction-of-arrival (DOA) estimation is a cornerstone of array signal processing with impact across wireless communications, radar, sonar, acoustics, and remote sensing [1], [2]. Arrays provide spatial filtering—i.e., the ability to separate signals that overlap in frequency but arrive from different directions [3]. This capability underpins systems from smart antennas and massive MIMO to passive sensing and localization, where accurate DOA translates directly into link reliability, interference suppression, tracking, and situational awareness.

A wide spectrum of methods has emerged to estimate DOA, which we group into two main families. **Classical beamformers** construct spatial filters to scan for power across angles, ranging from conventional (Bartlett) to adaptive

designs that suppress interference (e.g., Capon’s Minimum Variance Distortionless Response (MVDR)). The conventional Beamformer estimates power by steering and averaging; its angular resolution is fundamentally limited by the array’s beam pattern [4]. Capon’s high-resolution formulation improves upon this by designing a direction-selective, minimum-variance filter whose passband is fixed at the candidate angle while minimizing output power elsewhere [5]. **Subspace methods** recast DOA as a geometry problem on the array manifold. Multiple Signal Classification (MUSIC) estimates the signal and noise subspaces from the sample covariance and then searches for steering vectors orthogonal to the noise subspace [6]. Estimation of signal parameters via rotational invariant techniques (ESPRIT) removes the search by exploiting rotational (translational) invariance across paired subarrays. With a displacement-invariant geometry, the signal subspace obeys a simple eigen-relation whose eigenvalues encode inter-subarray phase shifts, and DOAs follow in closed form [7]. Unitary ESPRIT further constrains those phase factors to the unit circle and folds forward-backward averaging into a fully real-valued pipeline, yielding higher accuracy—especially for correlated sources—and lower computational burden [8], [9]. Root-MUSIC replaces spectral searches with polynomial rooting for Uniform Linear Array (ULA), improving resolution at a given Signal to Noise Ratio (SNR) [10].

Despite their elegance, classical approaches are sensitive to (i) limited snapshots (noisy covariance estimates), (ii) low SNR (weak subspace separability), (iii) multipath/coherent sources (rank deficiency), (iv) array/model mismatches such as gain/phase errors, position bias, and mutual coupling; remedies like spatial smoothing [11], [12], forward-backward averaging [13], and self calibration [14] help but are scenario-specific rather than universal. This tension motivates methods that restore the statistical structures classical algorithms rely upon, rather than replacing them.

The maturity of array processing theory affords rigorous lower bounds for benchmarking estimators. The most known among these is the Cramér–Rao lower bound (CRLB), which specifies the minimum achievable variance of any unbiased DOA estimator under the assumed data model and SNR, and thus serves as a canonical reference for asymptotic (high-SNR) performance [3], [15]. Complementing the CRLB, the Ziv–Zakai bound (ZZB) characterizes the transition to the threshold region by accounting for large-error events and closely spaced sources [16].

Over the past decade, deep learning has matured its able to

provide end-to-end systems that learn hierarchical representations and optimization objectives directly from data [17], [18]. In signal processing, model-based deep learning has bridged classical inductive biases with data-driven training, yielding architectures that encode domain structure while adapting to real-world deviations [19]. Within array processing and beamforming, recent surveys chart substantial gains across detection, localization, and interference suppression when learning is aligned with physics and deployment constraints [20].

The limitations that have been discussed above, have motivated data-driven alternatives. Recent learning-based DOA methods cluster into three lines. **Spectrum learning** trains a network to output a spatial spectrum that is later peak-searched. Early work learned spectra from engineered features and, when trained with designed imperfections, demonstrated robustness to gain/phase drift, position errors, and mutual coupling [21]. Follow-ups targeting low-SNR regimes showed that Convolutional Neural Networks (CNN) can sharpen peaks and suppress spurious responses compared with classical spectral baselines when operating conditions match training [22]. Other variants learn MUSIC-like spectra by classifying/aggregating responses over subregions [23], while snapshot-level models trained under joint imperfections keep spectrum formation outside the network (post-processing) [24]. These models are fast and simple to post-process, but their accuracy can degrade under train–test manifold mismatch and they may inherit on-grid/peak-association issues.

A second line keeps the physics and **learns the statistics** that classical methods need. *Covariance-reconstruction* networks denoise the sample covariance toward its ensemble counterpart. Closest to our philosophy, *SubspaceNet* ingests multi-lag autocorrelations, learns a covariance surrogate, and is trained *through Root-MUSIC*, so the signal/noise subspaces remain separable under coherence, limited data, and array mismatch—preserving interpretability and enabling plug-and-play with classical pipelines [25]. Related efforts such as Deep Augmented-Music (*DA-MUSIC*) learn priors for covariance (e.g., pseudo-covariance/rank structure) and subspace-guided spectral peaks, enabling robustness under coherence and broadband conditions while still exploiting the MUSIC model [26]. These hybrids typically generalize better across operating points and retain classical guarantees, at the cost of training complexity and continued reliance on (approximate) manifold knowledge.

Finally, **direct DOA predictors** output the angles themselves (classification or regression). CNNs operating on covariance-like “images” have reported strong performance in low-SNR conditions, albeit often for fixed small number of sources and with discretization effects when cast as classification [27]. A transformer variant, *DOA-ViT* which harness the Visual Transformer architecture (ViT), tokenizes spatial covariance matrices and uses supervised transfer learning to adapt from ideal to imperfect arrays with few labeled target samples—substantially improving robustness to array imperfections—though source coherence and cross-snapshot invariances are not explicitly modeled [28]. Relatedly, some hybrids first reconstruct the covariance and then attach a direct angle-prediction head,

combining statistical repair with end-to-end inference [29]. In short: spectrum learners emphasize speed and simple post-processing, hybrids *repair* the statistics that classical methods rely on, and direct predictors offer end-to-end simplicity; their trade-offs differ in handling variable number of sources, coherence, broadband effects, and array mismatch.

This work targets a persistent gap: a single, subspace-guided preprocessing model that reconstructs a clean covariance and its Eigen-Value-Decomposition (EVD) from noisy, imperfect measurements, so that Beamforming, MVDR, MUSIC, Root-MUSIC, and ESPRIT operate in their natural regime across low-SNR conditions, limited snapshots, coherent and non-coherent mixtures, and array imperfections—without per-scenario retuning. Following the “learn the statistics, keep the physics” principle, the network consumes a stack of multi-lag autocorrelations that is intrinsically informative for subspace recovery and less sensitive to sample length than raw snapshots. A U-Net backbone with skip connections [30], coupled to lightweight convolutional heads, predicts a Hermitian positive-definite covariance together with a clean eigendecomposition. Training couples a covariance-fidelity objective with a subspace-similarity objective, encouraging both statistical accuracy and preservation of the signal–noise geometry on which classical estimators rely. At inference, the reconstructed covariance replaces the sample estimate, and the predicted eigen-structure can be used directly when convenient; classical pipelines then run unchanged, retaining interpretability and avoiding on-grid biases typical of spectrum-classification networks.

We aim (i) to improve direction-finding accuracy at low SNR by restoring the eigen-gap; (ii) to remain robust under gain/phase drift, element-position bias, and mutual coupling; (iii) to handle both coherent and non-coherent scenes with a variable number of sources; and (iv) to show that a single trained model generalizes across operating regimes without case-by-case tuning. Evaluation compares against representative beamforming and subspace baselines under stresses commonly emphasized in covariance-reconstruction and hybrid-subspace studies (low SNR, limited data, source coherence, and array mismatch). The contributions of this work can outlined as follows:

- A preprocessing network that maps multi-lag autocorrelations to a clean covariance and eigendecomposition expressly to strengthen classical DOA estimators.
- A dual-objective training scheme that aligns covariance fidelity with subspace preservation, safeguarding the geometry exploited by MUSIC- and ESPRIT-type methods.
- A practical by-product in the form of a predicted eigendecomposition, which enables classical processing without online EVD and supports simple source-enumeration and confidence heuristics.

The remainder of this paper is organized as follows. Section II presents the array and signal models (ULA and imperfection assumptions) and states the DOA problem. Section III introduces the proposed subspace-guided preprocessing framework, including multi-lag autocorrelation features, the U-Net–based architecture, and the training objectives. Section IV details the experimental setup, datasets, baselines, and evaluation criteria.

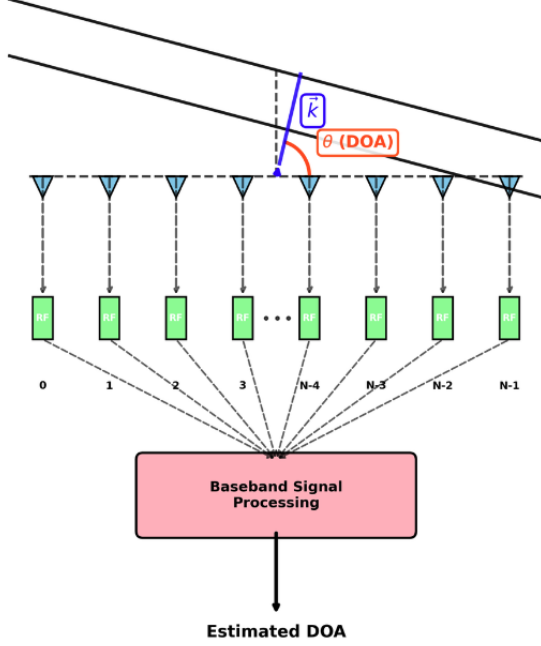


Fig. 1: Uniform linear array (ULA) geometry and angle convention. Element spacing is d , and the DoA θ is measured from the array axis (broadside = 90°).

Section V reports numerical results, highlighting accuracy, robustness, and generalization. Section VI concludes with a discussion on the contributions, limitations, and avenues for future work.

II. SIGNAL MODEL

Section outline. We first introduce the baseline (ideal) model in Section II-A, then formalize specular multipath/coherent sources in Section II-B, incorporate practical array imperfections in Section II-C, and finally state the problem formulation we solve (Section II-D). The array geometry and angle convention are illustrated in Figure 1.

A. Baseline (Ideal) Model

We consider an array of N sensors with inter-element spacing d . At snapshot $t \in \{1, \dots, T\}$, the array measurement is $\mathbf{x}(t) \in \mathbb{C}^{N \times 1}$ and the stacked data matrix is $\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(T)] \in \mathbb{C}^{N \times T}$. There are K emitters with directions of arrival (DoAs) $\boldsymbol{\theta} = [\theta_1, \dots, \theta_K]$, where θ is measured from the array axis (broadside = 90°). Let $\mathbf{s}(t) = [s_1(t), \dots, s_K(t)]^T \in \mathbb{C}^{K \times 1}$ denote the source vector and $\mathbf{n}(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I}_N)$ spatially white noise. The steering vector is

$$\mathbf{a}(\theta) = [e^{j\mathbf{k}(\theta)^T \mathbf{r}_0}, \dots, e^{j\mathbf{k}(\theta)^T \mathbf{r}_{N-1}}]^T \in \mathbb{C}^{N \times 1}, \quad (1)$$

$$\mathbf{k}(\theta) = \frac{2\pi}{\lambda} \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad \mathbf{r}_n \in \mathbb{R}^2. \quad (2)$$

and the steering matrix is

$$\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1), \dots, \mathbf{a}(\theta_K)] \in \mathbb{C}^{N \times K}. \quad (3)$$

For later use, the element-wise form is

$$[\mathbf{a}(\theta)]_n = e^{j\mathbf{k}(\theta)^T \mathbf{r}_n}, \quad n = 0, \dots, N-1, \quad (4)$$

Under linear superposition, the received signal at time t is

$$\mathbf{x}(t) = \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{n}(t). \quad (5)$$

Define the source and array covariances

$$\mathbf{R}_s \triangleq \mathbb{E}\{\mathbf{s}(t)\mathbf{s}^H(t)\} \succeq \mathbf{0}, \quad \mathbf{R}_x \triangleq \mathbb{E}\{\mathbf{x}(t)\mathbf{x}^H(t)\}, \quad (6)$$

so that

$$\mathbf{R}_x = \mathbf{A}(\boldsymbol{\theta}) \mathbf{R}_s \mathbf{A}^H(\boldsymbol{\theta}) + \sigma_n^2 \mathbf{I}_N, \quad (7)$$

and the sample covariance matrix (SCM)

$$\hat{\mathbf{R}}_x = \frac{1}{T} \mathbf{X} \mathbf{X}^H. \quad (8)$$

If $\text{rank}(\mathbf{R}_s) = K$ (mutually non-coherent sources), $\mathbf{R}_x$ splits into a K -dimensional signal subspace and an $(N - K)$ -dimensional noise subspace.

B. Specular Multipath / Coherent Sources

Specular multipath model: A dominant emitter can produce several mirror-like replicas of the line-of-sight (LoS) waveform via reflections. Each replica has excess path length $\Delta \ell_k$ expressed as group delay

$$\tau_k = \frac{\Delta \ell_k}{c}. \quad (9)$$

Let L_k be the path attenuation in dB, with linear amplitude gain

$$\alpha_k = 10^{-L_k/20}, \quad (10)$$

and let f_c denote the carrier frequency. With a phase offset ϕ_k which depends on material and polarization by the reflection coefficient phase, the complex-baseband per-path gain is

$$g_k = \alpha_k e^{j\phi_k} e^{-j2\pi f_c \tau_k}. \quad (11)$$

Correlation constraint on delay: Let B denote the essential baseband bandwidth of the direct signal $s_1(t)$ and $r_1(\Delta) = \mathbb{E}\{s_1(t)s_1^*(t - \Delta)\}$ its autocorrelation. The normalized envelope

$$\gamma(\tau) \triangleq \frac{|r_1(\tau)|}{r_1(0)} \quad (12)$$

decays on the correlation-time scale $T_{\text{coh}} \approx 1/B$. To remain in the highly correlated/specular regime,

$$0 \leq \tau_k \leq \tau_{\text{max}}, \quad \tau_{\text{max}} \approx \frac{1}{B}. \quad (13)$$

Received snapshot and covariance: Consider one direct source $s(t)$ from angle θ and one specular replica with parameters $(\tau, \theta^{\text{mp}}, g)$ that satisfy (9)–(11). Using the notation form (5), the array snapshot for single source is

$$\mathbf{x}(t) = \mathbf{a}(\theta) s(t) + \mathbf{a}(\theta^{\text{mp}}) g s(t - \tau) + \mathbf{n}(t). \quad (14)$$

Covariance expansion: Let $P \triangleq \mathbb{E}\{|s(t)|^2\}$ and $r(\Delta) \triangleq \mathbb{E}\{s(t)s^*(t-\Delta)\}$. The array covariance $\mathbf{R}_x = \mathbb{E}\{\mathbf{x}(t)\mathbf{x}^H(t)\}$ expands to

$$\begin{aligned} \mathbf{R}_x &= P \mathbf{a}(\theta) \mathbf{a}(\theta)^H \\ &+ |g|^2 P \mathbf{a}(\theta^{\text{mp}}) \mathbf{a}(\theta^{\text{mp}})^H \\ &+ g r(\tau) \mathbf{a}(\theta) \mathbf{a}(\theta^{\text{mp}})^H \\ &+ g^* r^*(\tau) \mathbf{a}(\theta^{\text{mp}}) \mathbf{a}(\theta)^H \\ &+ \sigma_n^2 \mathbf{I}_N. \end{aligned} \quad (15)$$

When $\tau < 1/B$ we typically have $|r(\tau)| \leq P$, so the replica is highly correlated with the direct path.

Rank-1 collapse and the effective steering vector: Well inside the correlation window write $r(\tau) = P\gamma$ with $|\gamma| \approx 1$ and $\gamma \approx e^{j\varphi}$. Denote $\mathbf{a}(\theta) \triangleq \mathbf{a}$ and $\mathbf{a}(\theta^{\text{mp}}) \triangleq \mathbf{b}$. The signal component of (15) (i.e., excluding $\sigma_n^2 \mathbf{I}_N$) is

$$\mathbf{R}_{x,\text{sig}} = P \mathbf{a} \mathbf{a}^H + |g|^2 P \mathbf{b} \mathbf{b}^H + g P \gamma \mathbf{a} \mathbf{b}^H + g^* P \gamma^* \mathbf{b} \mathbf{a}^H.$$

This factorizes as

$$\mathbf{R}_{x,\text{sig}} = P \left(\mathbf{a} + g \gamma \mathbf{b} \right) \left(\mathbf{a} + g \gamma \mathbf{b} \right)^H \triangleq P \mathbf{v}_{\text{eff}} \mathbf{v}_{\text{eff}}^H,$$

with the *effective steering vector*

$$\mathbf{v}_{\text{eff}} = \mathbf{a}(\theta) + g e^{j\varphi} \mathbf{a}(\theta^{\text{mp}}).$$

Thus, for coherent specular multipath ($|\gamma| = 1$), the signal covariance is exactly rank 1 (outer product). Intuitively, a narrowband delayed replica is a scaled, phase-rotated copy of the same waveform; there is only one independent temporal degree of freedom, so the spatial covariance is built from a single effective steering vector. For $|\gamma| \sim 1$, the matrix is *nearly* rank 1.

C. Array Imperfections

Geometry and nominal manifold: Nominal sensor locations are

$$\mathbf{r}_n^{(0)} = \begin{bmatrix} n \times d \\ 0 \end{bmatrix}, \quad n = 0, \dots, N-1, \quad (16)$$

with $k = \frac{2\pi}{\lambda}$ and $\mathbf{k}(\theta) = k [\cos \theta, \sin \theta]^T$. The nominal steering entry is

$$[\mathbf{a}_0(\theta)]_n = \exp(j \mathbf{k}(\theta)^T \mathbf{r}_n^{(0)}) = \exp(j \pi n \cos \theta). \quad (17)$$

Position errors (element misplacement): Each element is perturbed by

$$\boldsymbol{\varepsilon}_n \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}^2 \mathbf{I}_2), \quad (18)$$

giving actual position and position-perturbed manifold

$$\mathbf{r}_n = \mathbf{r}_n^{(0)} + \boldsymbol{\varepsilon}_n, \quad [\mathbf{a}_{\text{pos}}(\theta)]_n = \exp(j \mathbf{k}(\theta)^T \mathbf{r}_n). \quad (19)$$

This direction-dependent phase distortion induces steering-vector mismatch, DoA bias, and mainlobe broadening. *Configuration used in experiments:* $\sigma_{\text{pos}} = 0.001$ m.

Per-element gain/phase errors: Hardware variability which is representing a moderate quality mechanical design, fabrication and assembly procedures is modeled by a diagonal matrix

$$\begin{aligned} \mathbf{G} &= \text{diag}\{g_n e^{j(\pi/180) \phi_n}\}_{n=0}^{N-1}, \\ g_n &\sim \mathcal{U}[g_{\min}, g_{\max}], \quad \phi_n \sim \mathcal{U}[\phi_{\min}, \phi_{\max}] \text{ (degrees)}. \end{aligned} \quad (20)$$

introducing amplitude/phase imbalance and loss of nominal symmetry, where g_n and ϕ_n are uniformly distributed. *Configuration:* $g_n \in [0.975, 1.025]$; $\phi_n \in [-5^\circ, 5^\circ]$.

Mutual coupling (Toeplitz, reciprocal): Electromagnetic coupling due to moderate quality production procedure is captured by a Toeplitz matrix

$$\mathbf{M} = \mathbf{I}_N + \mathbf{E}, \quad \mathbf{E}[i, j] = \begin{cases} 0, & i = j, \\ c_{|i-j|}, & i \neq j, \end{cases} \quad (21)$$

with geometrically decaying off-diagonals

$$\begin{aligned} c_k &= \gamma^k v_k, \quad k = 1, \dots, N-1, \\ \gamma &= \rho e^{j\varphi_c}, \\ v_k &\sim \mathcal{U}[1 - \frac{\delta}{2}, 1 + \frac{\delta}{2}]. \end{aligned} \quad (22)$$

This reciprocal, distance-ordered model captures non-diagonal sensor interaction and loss of centro-symmetry relevant to Unitary-ESPRIT. *Configuration:* $\rho = 0.3$, $\varphi_c = -100^\circ$, $\delta = 0.9$.

Composite operator, distorted manifold, and covariance: Collecting hardware effects,

$$\begin{aligned} \mathbf{H} &= \mathbf{M} \mathbf{G}, \\ \tilde{\mathbf{a}}(\theta) &= \mathbf{H} \mathbf{a}_{\text{pos}}(\theta), \\ \tilde{\mathbf{A}}(\theta) &= [\tilde{\mathbf{a}}(\theta_1), \dots, \tilde{\mathbf{a}}(\theta_K)]. \end{aligned} \quad (23)$$

Component-wise,

$$[\tilde{\mathbf{a}}(\theta)]_i = \sum_{n=1}^N (\mathbf{M})_{i,n} g_n \exp(j(\pi/180)\phi_n) \exp(j \mathbf{k}(\theta)^T \mathbf{r}_n) \quad (24)$$

with $i = 0, \dots, N-1$.

The observation and covariance become the following.

$$\mathbf{x}(t) = \tilde{\mathbf{A}}(\theta) \mathbf{s}(t) + \mathbf{n}(t), \quad \mathbf{R}_x = \tilde{\mathbf{A}} \mathbf{R}_s \tilde{\mathbf{A}}^H + \sigma_n^2 \mathbf{I}_N. \quad (25)$$

Implications and severity knob:

- *Steering mismatch and bias:* position errors and gain/phase imbalance tilt $\tilde{\mathbf{a}}(\theta)$ away from $\mathbf{a}_0(\theta)$, biasing DoA estimates and degrading resolution.
- *Subspace leakage:* coupling mixes sensor outputs; even for uncorrelated sources, columns of $\tilde{\mathbf{A}}$ are not orthogonal under the noise metric, shrinking the signal-noise eigen-gap and flattening MUSIC-type spectra.
- *Loss of symmetry/centro-Hermiticity:* Toeplitz coupling and per-element phases break structures exploited by Unitary-ESPRIT and forward-backward averaging, weakening rotational invariance and increasing variance.

D. Problem Formulation

Objective.: Given T narrowband, far-field snapshots $\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(T)] \in \mathbb{C}^{N \times T}$ acquired by an array, estimate the set of directions of arrival (DoAs) $\boldsymbol{\theta} = \{\theta_1, \dots, \theta_K\}$. Depending on the setting, the source count K is either known (enumeration provided) or must be estimated jointly.

General model.: Snapshots obey the array models of Sections II-A to II-C. At time t ,

$$\mathbf{x}(t) = \tilde{\mathbf{A}}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{x}_{\text{mp}}(t) + \mathbf{n}(t),$$

where $\tilde{\mathbf{A}}(\boldsymbol{\theta})$ is the imperfection-distorted steering matrix (Section II-C), $\mathbf{x}_{\text{mp}}(t)$ is the contribution of coherent specular replicas (Section II-B), and $\mathbf{n}(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I}_N)$ is spatially white noise, independent of the signals. Signals may be mutually correlated or coherent. The sample covariance (SCM) and the ensemble covariance are

$$\hat{\mathbf{R}}_x = \frac{1}{T} \mathbf{X} \mathbf{X}^H, \quad \mathbf{R}_x = \mathbb{E}\{\mathbf{x}(t) \mathbf{x}^H(t)\}.$$

In the absence of coherence and with a matched manifold, $\mathbf{R}_x$ decomposes into a K -dimensional signal subspace and an $(N-K)$ -dimensional noise subspace; with highly correlated multipath (Section II-B) and imperfections (Section II-C), subspace leakage, rank collapse, and steering mismatch can occur.

Estimation task.: Let $\mathcal{G}$ denote a chosen decision rule (e.g., beamforming scan, MUSIC/Root-MUSIC spectrum, ESPRIT). The DOA estimator produces a set of angles $\hat{\boldsymbol{\theta}} = \mathcal{G}(\mathbf{X})$ (or $\mathcal{G}(\hat{\mathbf{R}}_x)$). Performance is evaluated with a permutation-invariant set error; writing Π_K for the set of $K \times K$ permutation matrices,

$$\text{RMSE}^2(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}) = \frac{1}{K} \min_{\mathbf{P} \in \Pi_K} \|\hat{\boldsymbol{\theta}} - \mathbf{P} \boldsymbol{\theta}\|_2^2,$$

and we report RMSE in degrees over test scenes.

Operating conditions.: We are interested in regimes with low SNR, limited T , possible source coherence (specular replicas with small delays), and realistic array imperfections. These conditions degrade eigengaps and peak separability and thus constitute the stressors against which estimators are assessed in Sections IV and V.

III. PROPOSED METHOD

A. Overview

We propose a subspace-guided preprocessing network that maps raw snapshots to two physics-consistent products: (i) a denoised, Hermitian covariance (“clean SCM”), and (ii) a self-consistent eigen-decomposition (“clean EVD”), from now on let us use the name ReconUNet when referring to our proposed method for its ability to reconstruct the clean covariance matrix and for its usage of the UNet architecture. The network *does not* replace classical DoA estimators; instead, it repairs second-order structure so that MUSIC/Root-MUSIC/ESPRIT etc, operate closer to their ideal regime under low SNR, coherent multipath, and array imperfections. See Figure 2 for the training flow and Figure 3 for the inference pipeline.

B. Feature Representation: Multi-Lag Autocorrelation

Given snapshots $\{\mathbf{x}(t)\}_{t=0}^{T-1}$, we form short-range temporal autocorrelations up to lag τ :

$$\begin{aligned} \hat{\mathbf{R}}_x[\ell] &= \frac{1}{T-\ell} \sum_{t=\ell+1}^T \mathbf{x}(t) \mathbf{x}^H(t-\ell), \\ \hat{\mathbf{R}}_x[\ell] &\in \mathbb{C}^{N \times N}, \quad \ell = 0, \dots, \tau-1. \end{aligned} \quad (26)$$

We stack the τ complex matrices along a channel axis and convert to a real tensor via

$$\begin{aligned} \text{Stack}_{\Re\Im} : \mathbb{C}^{N \times N} &\rightarrow \mathbb{R}^{2N \times N}, \\ \mathbf{Z} &\mapsto \begin{bmatrix} \Re\{\mathbf{Z}\} \\ \Im\{\mathbf{Z}\} \end{bmatrix}, \end{aligned} \quad (27)$$

so the network input has shape $B \times \tau \times 2N \times N$ (B -batch).

Why this feature works:

- *Lower variance, same physics:* second-order statistics concentrate what subspace methods use (SCM/EVD) and average temporal noise.
- *Multipath cues:* correlated replicas that collapse rank at $\ell=0$ leave structure at small $\ell>0$, exposed by stacking lags.
- *Useful invariances:* autocorrelation removes absolute phase/time shifts; unit-norm steering yields stable dynamic range.
- *Short lags suffice:* small τ captures informative temporal structure without inflating input size.

C. Network Architecture

U-Net backbone: A compact 2D U-Net processes the lag-sensor grid with three encoder/decoder levels. Each encoder stage applies two 3×3 convolutions (with normalization and nonlinearity) and 2×2 downsampling; the decoder mirrors this with transposed convolutions and skip connections that preserve fine diagonal/Hermitian patterns. *Rationale.* U-Nets are a natural fit for image-to-image translation, and the lag-sensor tensor is an image whose “textures” are the algebraic regularities of array covariance: narrow diagonals encode per-sensor power, near-diagonals encode short-baseline correlations, and longer-range structure encodes mutual coupling and cross-lag correlations. The encoder’s growing receptive field aggregates global context across *lags* (capturing coherence/multipath signatures and rank structure) and across *sensors* (capturing array-wide distortions such as coupling and misplacement), while the decoder reconstructs a full matrix with physically consistent local detail. Skip connections act as an inductive bias for preserving fine-grained conjugate/Hermitian patterns that would otherwise be washed out by downsampling—critical for accurate eigengaps and subspace geometry. Practically, this architecture is parameter-efficient, translation-equivariant on the grid (useful for stationarity along sensor and lag axes), and empirically strong for denoising/inpainting tasks, which matches our goal of repairing second-order statistics rather than regressing angles directly.

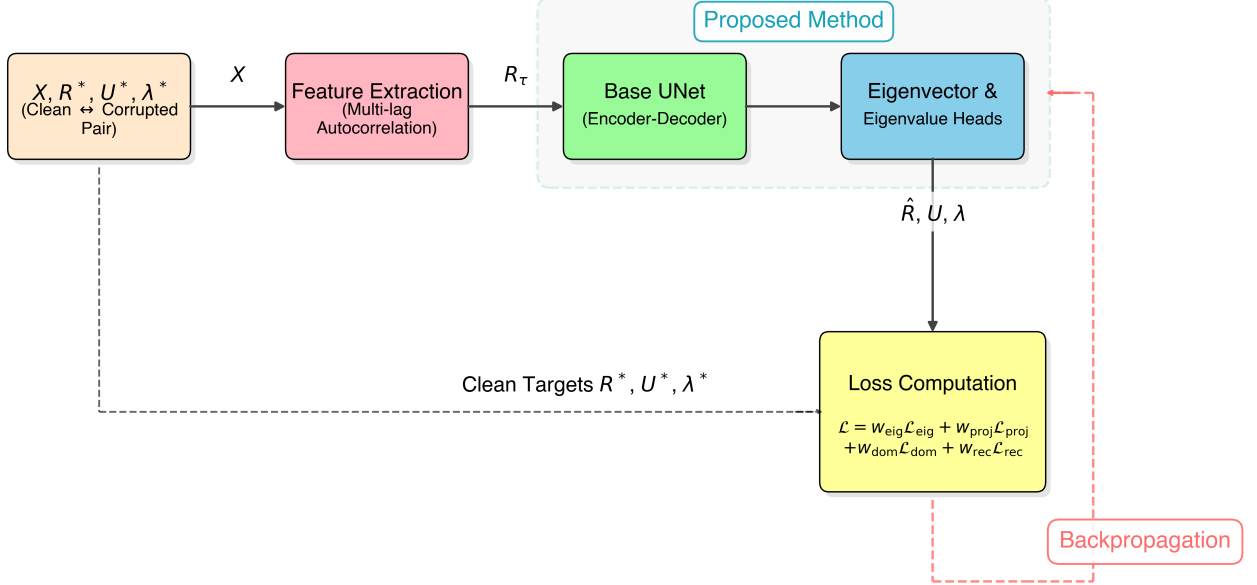


Fig. 2: Training pipeline: from corrupted snapshots to lag-stack features, U-Net heads, and loss computation.

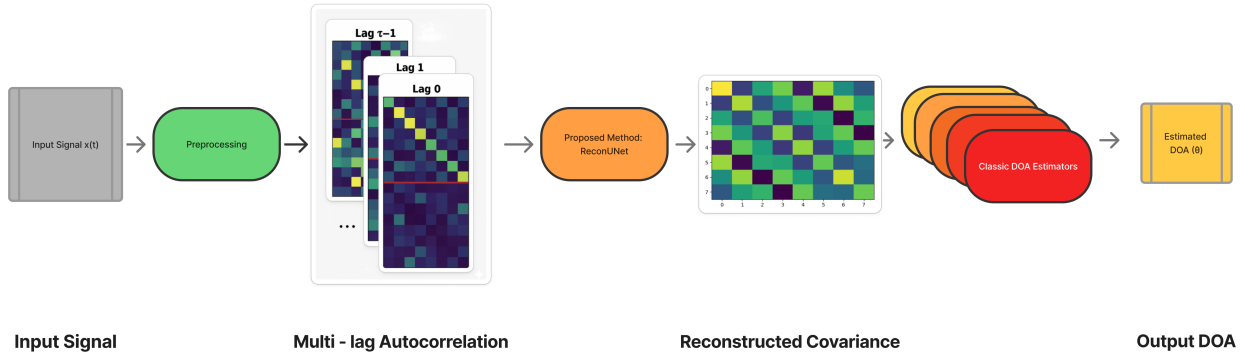


Fig. 3: End-to-end inference pipeline (multilag autocorrelation → EVD U-Net → classical DoA estimators).

Anti-rectifier activation (SubspaceNet-style): Besides standard ReLU, we employ an *anti-rectifier* that concatenates positive and negative responses to preserve sign information critical for real/imag structure:

$$\text{AR}(\mathbf{X}) = [\text{ReLU}(\mathbf{X}) \parallel \text{ReLU}(-\mathbf{X})] \quad (28)$$

(concatenation along the channel axis). This yields phase-complementary channels, improves gradient flow near zero, and better models skew-symmetry in the imaginary plane. The resulting architecture is illustrated in Figure 4.

D. SCM Head: Hermitian Projection and Diagonal Loading

The decoder predicts an $N \times N$ complex matrix via the inverse of (27). We enforce Hermitianity and stabilize conditioning:

$$\hat{\mathbf{R}}_{\text{Herm}} = \frac{1}{2}(\hat{\mathbf{R}}_{\text{raw}} + \hat{\mathbf{R}}_{\text{raw}}^H), \quad \hat{\mathbf{R}} = \hat{\mathbf{R}}_{\text{Herm}} + \varepsilon \mathbf{I}_N, \quad (29)$$

with a small load $\varepsilon > 0$.

- *Hermitian projection* maps to the physically admissible set of array covariances.

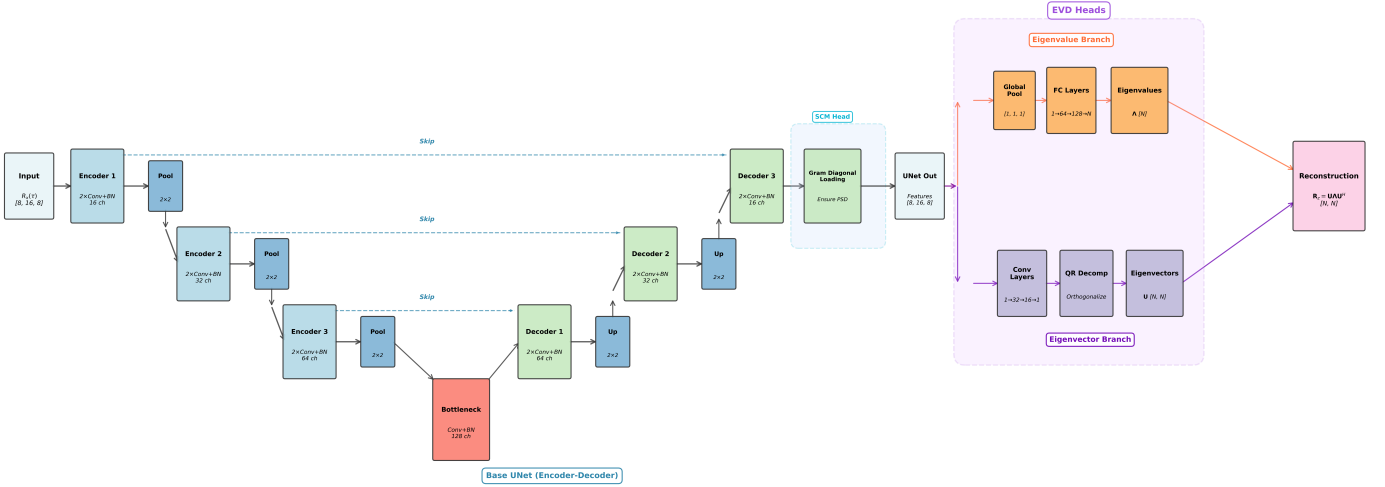


Fig. 4: U-Net backbone with skip connections with SCM head followed by the EVD heads.

- *Gram diagonal overloading* (Tikhonov-style) improves PSD-ness and eigengap clarity, stabilizing downstream EVD/subspace steps.

E. EVD Heads: Spectrum and Basis with Complex QR

Eigenvalue head: Global average pooling over the $(2N \times N)$ plane followed by an MLP predicts nonnegative eigenvalues $\hat{\lambda} \in \mathbb{R}_+^N$; we sort them in descending order using a permutation Π :

$$\hat{\lambda}_\downarrow = \Pi \hat{\lambda}, \quad \hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_N. \quad (30)$$

Eigenvector head with complex QR and phase anchoring: A shallow CNN predicts $\hat{\mathbf{U}}_{\text{raw}} \in \mathbb{C}^{N \times N}$ (real/imag stack $\rightarrow$ complex). We project to the Stiefel manifold via a differentiable complex QR:

$$[\hat{\mathbf{Q}}, \hat{\mathbf{R}}] = \text{cQR}(\hat{\mathbf{U}}_{\text{raw}}), \quad \hat{\mathbf{U}} = \hat{\mathbf{Q}}, \quad \hat{\mathbf{U}}^H \hat{\mathbf{U}} = \mathbf{I}_N. \quad (31)$$

To resolve per-column complex phase, we *anchor* each column by rotating a reference entry (e.g., first row) to real-positive:

$$\varphi_i = \arg([\hat{\mathbf{U}}]_{1,i}), \quad \bar{\mathbf{U}} = \hat{\mathbf{U}} \text{diag}(e^{-j\varphi_1}, \dots, e^{-j\varphi_N}). \quad (32)$$

The EVD-reconstructed covariance used for supervision and inference is

$$\hat{\mathbf{R}}_{\text{EVD}} = \bar{\mathbf{U}} \text{diag}(\hat{\lambda}_\downarrow) \bar{\mathbf{U}}^H. \quad (33)$$

F. Training Data Generation and Objectives

Inputs (corrupted): For each scene we synthesize snapshots and features under the imperfect array of Section II-C, with *per-scene* randomization of (i) per-element gain/phase errors $\mathbf{G}$, (ii) small position jitter $\{\varepsilon_m\}$, and (iii) reciprocal Toeplitz mutual coupling $\mathbf{M}$. Optional correlated replicas (specular multipath) follow Section II-B. Additive noise is drawn to meet the target SNR. These corrupted snapshots are converted into the multi-lag autocorrelation stack used as network input (cf. Section III-B).

Targets (clean, ideal-manifold): Supervision is provided by a *clean*, imperfection-free covariance and its eigendecomposition, computed on the nominal array manifold of Section II-A with *no array imperfections and no multipath*. Let θ denote the direct-path DoAs in the scene, $\mathbf{A}_0(\theta)$ the nominal steering matrix, and $\mathbf{R}_s^* \succeq \mathbf{0}$ the source covariance (diagonal for non-coherent sources). The target covariance is

$$\mathbf{R}^* = \mathbf{A}_0(\theta) \mathbf{R}_s^* \mathbf{A}_0^H(\theta) \quad (34)$$

Its eigendecomposition is $\mathbf{R}^* = \mathbf{U}^* \text{diag}(\lambda^*) (\mathbf{U}^*)^H$, with the first K columns of $\mathbf{U}^*$ spanning the signal subspace and the remaining $N-K$ columns spanning the noise subspace.

Model outputs: Given the corrupted inputs, the network predicts an orthonormal basis $\hat{\mathbf{U}} \in \mathbb{C}^{N \times N}$ (obtained via a differentiable complex QR projection), a nonnegative spectrum $\hat{\lambda} \in \mathbb{R}_+^N$ (sorted descending), and the reconstructed covariance

$$\hat{\mathbf{R}} = \hat{\mathbf{U}} \text{diag}(\hat{\lambda}) \hat{\mathbf{U}}^H. \quad (35)$$

Training objective: We minimize a composite loss that ties the predicted objects to their clean counterparts:

$$\mathcal{L} = w_{\text{eig}} \mathcal{L}_{\text{eig}} + w_{\text{proj}} \mathcal{L}_{\text{proj}} + w_{\text{dom}} \mathcal{L}_{\text{dom}} + w_{\text{rec}} \mathcal{L}_{\text{rec}}, \quad (36)$$

with hyper-parameter weights $(w_{\text{eig}}, w_{\text{proj}}, w_{\text{dom}}, w_{\text{rec}})$. Column orthonormality is enforced by the QR layer, so no explicit orthogonality penalty is used.

(1) *Signal-subspace alignment (projection-Frobenius).* For a sample with K sources, let $\mathbf{U}_K^* \in \mathbb{C}^{N \times K}$ and $\hat{\mathbf{U}}_K \in \mathbb{C}^{N \times K}$ be the leading K columns of $\mathbf{U}^*$ and $\hat{\mathbf{U}}$. The Grassmannian (projection) loss is

$$\begin{aligned} \mathcal{L}_{\text{proj}} &= \left\| \hat{\mathbf{U}}_K \hat{\mathbf{U}}_K^H - \mathbf{U}_K^* (\mathbf{U}_K^*)^H \right\|_F^2 \\ &= 2K - 2 \left\| (\mathbf{U}_K^*)^H \hat{\mathbf{U}}_K \right\|_F^2, \end{aligned} \quad (37)$$

which is invariant to per-column phase and basis rotations within the subspace.

(2) *Dominant-eigenvector tie (sign-robust).* For $K \geq 1$, with u_1^* and $\hat{u}_1$ the leading columns of $\mathbf{U}^*$ and $\hat{\mathbf{U}}$,

$$\mathcal{L}_{\text{dom}} = \min \left(\|\hat{u}_1 - u_1^*\|_2^2, \|\hat{u}_1 + u_1^*\|_2^2 \right), \quad (38)$$

Algorithm 1: Training using subspace-guided statistics repair (EVD-only head)

Init: Lags τ ; loss weights $(w_{\text{eig}}, w_{\text{proj}}, w_{\text{dom}}, w_{\text{rec}})$; learning rate μ ; epochs e_{max} ; batches per epoch B .

Input: Training set $\mathcal{D} = \{(\mathbf{X}, \mathbf{R}^*, \mathbf{U}^*, \boldsymbol{\lambda}^*, K)\}$, where $\mathbf{X} \in \mathbb{R}^{(\tau+1) \times (2N) \times N}$ is the stored lag-stack tensor.

- 1 Initialize network parameters Θ .
- 2 **for** $epoch \leftarrow 1$ **to** e_{max} **do**
- 3 Randomly partition $\mathcal{D}$ into B mini-batches $\{\mathcal{D}_b\}_{b=1}^B$.
- 4 **for** $b \leftarrow 1$ **to** B **do**
- 5 Load batch $\{(\mathbf{X}, \mathbf{R}^*, \mathbf{U}^*, \boldsymbol{\lambda}^*, K)\} \in \mathcal{D}_b$.
- 6 **Forward:**
- 7 $(\hat{\mathbf{U}}_{\text{raw}}, \hat{\boldsymbol{\lambda}}) \leftarrow f_{\Theta}(\mathbf{X})$.
- 8 **Unitary projection & sorting:**
- 9 $\hat{\mathbf{U}} \leftarrow \text{cQR}(\hat{\mathbf{U}}_{\text{raw}})$; sort $\hat{\boldsymbol{\lambda}}$ in descending order.
- 10 **Phase anchoring:**
- 11 Rotate columns of $\hat{\mathbf{U}}$ so a chosen reference entry per column is real-positive $\Rightarrow \bar{\mathbf{U}}$.
- 12 **Recompose covariance:**
- 13 $\hat{\mathbf{R}} = \bar{\mathbf{U}} \text{diag}(\hat{\boldsymbol{\lambda}}) \bar{\mathbf{U}}^H$. (33)
- 14 **Losses:**
- 15 $\mathcal{L}_{\text{proj}}$ (37), $\mathcal{L}_{\text{dom}}$ (38), $\mathcal{L}_{\text{eig}}$ (39), $\mathcal{L}_{\text{rec}}$ (40).
- 16 **Aggregate:**
- 17 $\mathcal{L} = w_{\text{proj}}\mathcal{L}_{\text{proj}} + w_{\text{dom}}\mathcal{L}_{\text{dom}} + w_{\text{eig}}\mathcal{L}_{\text{eig}} + w_{\text{rec}}\mathcal{L}_{\text{rec}}$. (36)
- 18 **Update:**
- 19 $\Theta \leftarrow \Theta - \mu \nabla_{\Theta} \mathcal{L}$ (optimizer step, e.g., Adam).
- 20 **return** Θ

preserves fine per-element structure while being insensitive to a global sign.

(3) *Eigenvalue fidelity.*

$$\mathcal{L}_{\text{eig}} = \frac{1}{N} \|\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}^*\|_2^2, \quad (39)$$

stabilizes eigengaps and scales.

(4) *EVD consistency (reconstruction tie).*

$$\mathcal{L}_{\text{rec}} = \frac{1}{N^2} \|\hat{\mathbf{R}} - \mathbf{R}^*\|_F^2 = \frac{1}{N^2} \sum_{i,j} |\hat{R}_{ij} - R_{ij}^*|^2. \quad (40)$$

Together, $\mathcal{L}_{\text{proj}}$ and $\mathcal{L}_{\text{dom}}$ encourage accurate *signal* subspaces, $\mathcal{L}_{\text{eig}}$ preserves spectral structure and eigengaps, and $\mathcal{L}_{\text{rec}}$ enforces that the predicted eigenpairs recompose a valid clean covariance on the ideal manifold per (34).

G. Inference and Classical Integration

At test time, we compute (26), pass the stack through the network, and proceed via either route:

Algorithm 2: Inference using subspace-guided statistics repair

Init: Trained network parameters Θ ; known source count K ; angle grid $\Theta_{\text{grid}} = \{0^\circ:1^\circ:180^\circ\}$ with parabolic refinement; chosen classical DoA estimator.

Input: Lag-stack tensor $\mathbf{X} \in \mathbb{R}^{\tau \times 2N \times N}$ formed from snapshots via (26)–(27).

- 1 Apply network to $\mathbf{X}$ to obtain EVD outputs $(\hat{\mathbf{R}}, \bar{\mathbf{U}}, \hat{\boldsymbol{\lambda}}) \leftarrow f_{\Theta}(\mathbf{X})$.
- 2 Apply the chosen estimator using one of the routes:
- 3 *Subspace route:* use $\bar{\mathbf{U}}$ to form signal/noise bases and run MUSIC/Root-MUSIC/ESPRIT to get $\hat{\boldsymbol{\theta}}$.
- 4 *Reconstructed Covariance route:* use $\hat{\mathbf{R}}$ directly inside the estimator; for Bartlett/MVDR/MUSIC, scan Θ_{grid} and refine peaks.
- 5 **return** $\hat{\boldsymbol{\theta}}$.

- *Covariance route:* feed $\hat{\mathbf{R}}_{\text{EVD}}$ from (33) to MUSIC/Root-MUSIC/ESPRIT (with K provided or estimated externally).
- *Subspace route:* use columns of $\bar{\mathbf{U}}$ as the signal/noise basis, bypassing runtime EVD.

Angle extraction remains purely classical; the network focuses on repairing second-order structure.

The inference pipeline is detailed in Alg. 2.

IV. TRAINING APPROACH AND EXPERIMENTAL SETUP

Scope. We state the data synthesis, input features, supervision signals, exact training hyperparameters, baselines, metrics, and stress scenarios required to reproduce the results. Architectural details and loss definitions appear earlier.

Signal model and geometry. Unless noted otherwise: ULA with $N=8$ and half-wavelength spacing $d=\lambda/2$, narrowband around 2.45 GHz, array-axis angle convention $\theta \in [0^\circ, 180^\circ]$ (broadside = 90°), and $T=512$ snapshots per scene. SNR is defined w.r.t. the *main* source power P as $\text{SNR} = 10 \log_{10}(P/\sigma_n^2)$; σ_n^2 is set from the target. We restrict sampling and scanning to $[30^\circ, 150^\circ]$ to avoid endfire ambiguities and poor conditioning at $\{0^\circ, 180^\circ\}$ where a ULA's cosine manifold is least informative.

Dataset generation (paired *clean* $\leftrightarrow$ *corrupted*). Learning is cast as *statistics repair*. For each scene we store:

- *Target (clean).* Ideal manifold, no multipath, no imperfections; the clean covariance $\mathbf{R}^*$ and its EVD $(\mathbf{U}^*, \boldsymbol{\lambda}^*)$ serve as supervision.
- *Input (corrupted).* The same scene rendered with additive noise at the chosen SNR, optional *correlated* specular replicas, and array imperfections (per-element gain/phase offsets, small position jitter, reciprocal Toeplitz mutual coupling) per Section II-C.

Angles and separation: direct-path DoAs are drawn from $\{30^\circ:150^\circ\}$ with a minimum 10° separation between *direct* sources; replicas are unconstrained. *Counts:* one main source plus $I \in \{0, 1, 2, 3\}$ equal-power interferers (so SIR = 0 dB when $I>0$), giving $K = 1 + I$. *correlated multipath:* $M \in \{0, 1, 2, 3\}$ narrowband replicas with small delays and

independent departure angles. We render both perfect and imperfect arrays, with imperfection parameters randomized per scene.

Dataset size and splits. The corpus comprises $\approx 2.0 \times 10^6$ paired examples (corrupted input $\rightarrow$ clean target), covering both perfect and imperfect arrays across the SNR schedule below. Train/val/test partitions are fixed by seed; all reported numbers in Section V use the held-out test split.

Feature construction (network inputs). From corrupted snapshots we compute short-lag autocorrelations

$$\hat{\mathbf{R}}_x[\ell] = \frac{1}{T-\ell} \sum_{t=\ell+1}^T \mathbf{x}(t) \mathbf{x}^H(t-\ell), \quad \ell = 0, \dots, \tau-1 \quad (41)$$

then split real/imag to form a real tensor of shape $B \times \tau \times 2N \times N$.

Supervision and losses. Each sample provides its ground-truth K (direct-sources count) to index the signal subspace as in Section III-F. We use the composite objective in (36): (i) projection–Frobenius subspace alignment (37), (ii) dominant-eigenvector tie (38), (iii) eigenvalue fidelity (39), and (iv) EVD consistency (40). Column orthonormality is enforced by the differentiable complex-QR layer; an orthogonality metric is monitored but not penalized.

Exact training hyperparameters. Unless stated otherwise, the following settings are used throughout:

- *Model configuration:* temporal lag depth $\tau=8$; number of sensors $N=8$ (input grid uses $2N \times N$ via real/imag split); activation type = anti-rectifier; dropout enabled. The SCM head performs Hermitian symmetrization and adds a diagonal load $\varepsilon=1$ (cf. Section III-D).
- *Optimizer:* Adam, learning rate 1×10^{-4} , weight decay 1×10^{-5} .
- *Scheduler:* ReduceLROnPlateau on validation loss (mode=min), factor 0.7, patience 5 epochs.
- *Batching:* batch size 2048.
- *Epochs and early stopping:* up to 300 epochs; early-stopping patience 25 epochs on validation loss.
- *Gradient clipping:* global ℓ_2 -norm clipped at 1.0.
- *SNR schedule:* training scenes sweep SNR $[-10, 10]$ dB; evaluation sweeps $[-20, 20]$ dB in the scenario study.
- *Imperfection randomization:* per scene as in Section II-C.

Training dynamics. Figure 5 summarizes the evolution of the total objective on train/validation splits across epochs.

Inference and classical integration. At test time we form the lag-stack, apply the network, and proceed via either route:

- *Reconstructed Covariance route:* feed the reconstructed covariance $\hat{\mathbf{R}}_{\text{clean}}$.
- *Subspace route:* use the predicted eigenbasis to form signal/noise subspaces, bypassing runtime EVD.

Angle extraction remains purely classical; the learned front-end only repairs second-order structure.

Baselines and configurations. We evaluate each estimator *as-is* on (i) the raw SCM $\hat{\mathbf{R}}_x$ and (ii) our reconstructed covariance $\hat{\mathbf{R}}_{\text{clean}}$:

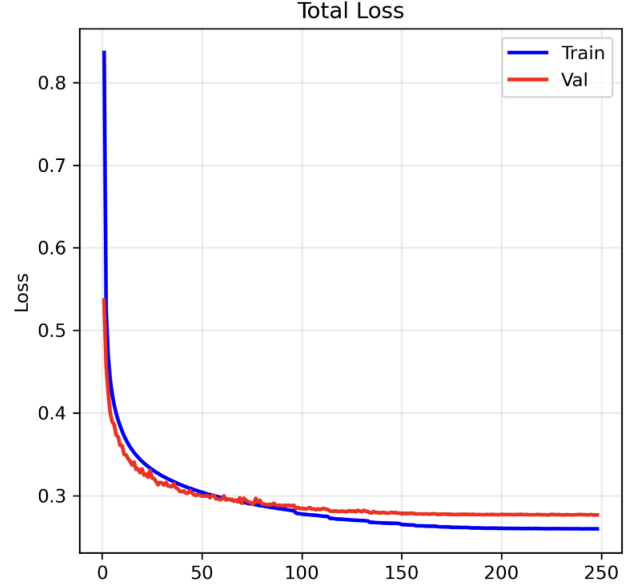


Fig. 5: Total loss over epochs on train/validation splits.

- Conventional (Bartlett) and MVDR (Capon) beamformers,
- MUSIC with known K ; Root-MUSIC (ULA),
- ESPRIT and Unitary-ESPRIT.

Spectrum scans use a 1° grid on $[30^\circ, 150^\circ]$ with three-point parabolic refinement around peaks for sub-degree accuracy. In all cases K is assumed known and identical across conditions.

Evaluation protocol and metrics. Accuracy is reported as RMSE of angular error (degrees) after optimal assignment between the K estimated angles and ground truth. Plots typically employ a logarithmic y -axis to emphasize low-error regimes; where informative, we overlay the Cramér–Rao bound.

Qualitative diagnostics. In addition to RMSE, we report two diagnostic visualizations in Section V. (i) A *trajectory consistency plot* that sweeps three ground-truth angles across a sequence of samples at fixed $K=3$ and SNR=0dB with a constant separation equals 10° between sources, and overlays the corresponding estimates from each subspace method (MUSIC/Root-MUSIC/ESPRIT). This reveals alignment and bias across a controlled angular sweep. (ii) A *spectral comparison* that contrasts spectra computed from the noisy SCM and from the reconstructed covariance $\hat{\mathbf{R}}_{\text{clean}}$ (e.g., MUSIC pseudospectra), highlighting peak sharpening and spurious-peak suppression attributable to the learned statistics repair.

Scenario matrix (stress tests). Each scenario is evaluated under *perfect* and *imperfect* arrays while sweeping SNR $\in [-20, 20]$ dB:

- Basic: one direct source (no multipath).
- Moderate: two direct sources plus one coherent replica (of one source).
- Advanced 1: one direct source with six coherent replicas (a scenario which was not part of the training grid where $M \leq 3$).
- Advanced 2: four direct sources with three coherent replicas spread across emitters.

The 10° minimum-separation rule applies only to direct sources; coherent replicas are unconstrained.

V. RESULTS AND DISCUSSION

Scope. We evaluate the estimators and scenarios defined in Section IV and report accuracy as RMSE of angular error (degrees). We refrain from restating the setup and refer the reader to Section IV for geometry, data generation, feature construction, baselines, and metrics.

A. Ideal-Case Baseline: Classical Methods Under Matched Models

Objective. Before introducing our front-end, we establish the performance of classical estimators when the data exactly satisfy their assumptions (Basic scenario in Section IV: narrowband, no multipath, no array imperfections, known K).

Findings. In Figure 6 we report *only* the classical estimators using the same 1° scan grid adopted throughout the paper for consistency. Under matched conditions, Root-MUSIC and (Unitary-)ESPRIT track the CRLB closely. By contrast, the *grid-scanning* methods (MUSIC, MVDR) exhibit a characteristic high-SNR flattening: as SNR increases, their spectral peaks sharpen and the combination of a discrete angle grid with three-point parabolic refinement imposes a finite angular resolution, yielding an apparent performance floor. MVDR experiences the same effect because its mainlobe narrows with SNR, amplifying sub-grid localization limits; Bartlett appears less affected since its broader peaks make the parabolic fit less sensitive to discretization. The *gridless* methods (Root-MUSIC, ESPRIT/Unitary-ESPRIT) avoid discretization entirely and therefore do not exhibit this high-SNR floor.

For completeness, we also evaluated MUSIC/MVDR with a finer 0.1° grid. As expected, this reduces discretization error. Root-MUSIC/ESPRIT curves are essentially unchanged under finer grids, as they do not rely on scanning. The finer grid increases scan cost by roughly $\times 10$ for a fixed field of view, which is a practical trade-off when targeting sub-tenth-degree accuracy in high-SNR, well-modeled conditions. Representative 0.1° results are shown in Figure 7, and pivot statistics appear in Table I for the 1° setting.

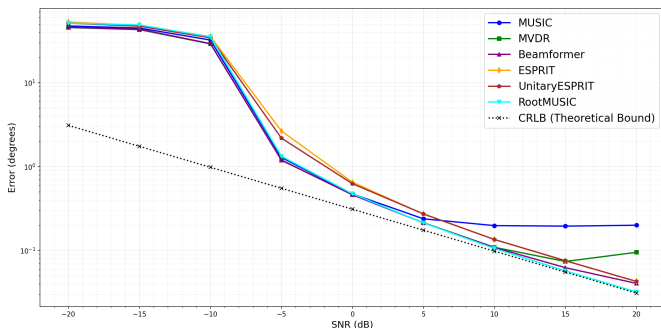


Fig. 6: Ideal-case benchmark (Basic scenario, no multipath, no imperfections): RMSE vs. SNR for classical estimators with CRLB overlay.

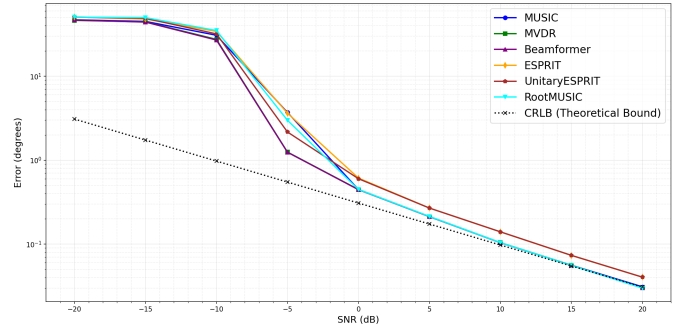


Fig. 7: Ideal-case benchmark with a finer 0.1° scan grid: RMSE vs. SNR, with CRLB overlay.

TABLE I: Ideal-case RMSE (deg) at Low (-10 dB), Mid (5 dB), and High (20 dB) SNR. Best per column in **bold**.

Method	Low SNR	Mid SNR	High SNR
Bartlett	27.3631	0.2071	0.0379
MVDR	26.8068	0.2080	0.0915
MUSIC	33.6959	0.2360	0.1971
Root-MUSIC	37.2366	0.2070	0.0316
(Unitary-)ESPRIT	35.7450	0.2758	0.0408

Interpretation. This baseline confirms that when assumptions match reality, classical subspace methods are already near-optimal; our learned front-end is aimed at restoring *similar* behavior in the stressed regimes of Section IV.

B. RMSE vs. SNR Across Scenarios

Objective. Using the scenarios defined in Section IV, we compare each classical estimator fed with (i) the raw SCM and (ii) our reconstructed covariance / predicted EVD, across an SNR sweep. To demonstrate *plug-and-play* compatibility, we deliberately vary the classical *subspace* back end across scenarios (MUSIC, ESPRIT, and Unitary-ESPRIT). The last panel stresses *out-of-distribution* (OOD) generalization with a single direct source and *six* coherent replicas—more multipath than seen during training.

Evaluation set construction (consistent angles across SNR). To make the SNR sweeps directly comparable, we fixed the scene geometry across SNR levels and only changed the noise power. Concretely, the evaluation set comprises 9,000 total samples arranged as:

- **9 SNR levels:** $-20, -15, -10, -5, 0, 5, 10, 15, 20$ dB;
- **1,000 samples per SNR level** (same angles at every SNR).

Evaluation dataset creation process.

- 1) **I):** draw 1,000 unique angle configurations (numbers of sources, their DoAs, and any multipath layout) according to the scenario definition. These indices are fixed and reused for all SNRs.
- 2) **II (per SNR):** for each of the same 1,000 angle configurations, render one sample at each of the 9 SNR levels by adjusting only σ_n^2 . Thus, *the same angles appear at every SNR*.

Findings. Observations from Figs. 8–10:

- **Consistency across subspace methods.** In each panel we intentionally pair our front end with a *different* classical subspace algorithm (MUSIC in Figure 10, ESPRIT in Figure 8, Unitary-ESPRIT in Figure 9). The dashed curves (UNet_MUSIC, UNet_ESPRIT, UNet_UNITARY_ESPRIT) improve upon their solid counterparts over wide SNR ranges, confirming that the proposed statistics repair is *algorithm-agnostic* and integrates seamlessly with distinct subspace engines.
- **Low-SNR and coherence regimes.** The largest gains appear where the signal-noise eigengap is smallest (low SNR and coherent mixtures).
- **Crowded scenes.** In the 4+3 case (Figure 9) the dashed Unitary-ESPRIT curve remains well below its solid baseline across the sweep, indicating that the front end suppresses spurious peaks and improves assignment stability even under heavy overlap.
- **Out-of-distribution behavior at high SNR.** In the OOD setting with 1 direct source and 6 coherent replicas (Figure 10), our method markedly improves RMSE in the low-to-mid SNR range where it was trained to operate; however, for SNRs *above* roughly 10 dB, the dashed UNet_MUSIC curve turns upward while the best classical baselines continue to decrease. This suggests a *regularization-induced floor*: the network, optimized primarily for low-SNR robustness and trained within a narrower SNR window, may over-smooth or bias the statistics in clean regimes it did not see during training. Practically, this indicates that the front end is best used where covariance estimates are noisy or ill-conditioned; at very high SNR in strongly off-distribution scenes, direct classical processing may be preferable unless the training schedule and losses are adapted for that regime.

Numerical summaries at 0 dB SNR are reported in Table II. As evident in Table II, the ESPRIT-based variant achieves the lowest RMSE across scenarios. This is expected: ESPRIT operates on the *signal* subspace—the very object our network is trained to reconstruct and align—whereas MUSIC relies on the *noise* subspace and a discrete grid, making it more sensitive to residual leakage and discretization error. In the crowded case, ESPRIT also outperforms Unitary-ESPRIT because the latter depends on forward-backward symmetry (centro-Hermiticity); array imperfections and coupling violate this structural assumption, degrading Unitary-ESPRIT more than standard ESPRIT.

C. Angle-Sweep Performance (3 Sources, No Multipath)

Setup. We sweep three ground-truth angles across the field of view at fixed separation $\Delta\theta = 10^\circ$ and 0 dB SNR, with no multipath and an imperfect array (see Section IV). For each method we overlay ground truth (solid lines) and the tracked estimates (markers) versus sample index. We report classical MUSIC, Root-MUSIC, and ESPRIT, and the same back ends when fed by our reconstructed covariance / predicted EVD. **Method-wise observations at 0 dB, $K=3$, $\Delta\theta = 10^\circ$.** Beyond the aggregate trend, the six panels in Figure 11 reveal characteristic behaviors for each classical back end and its aided counterpart:

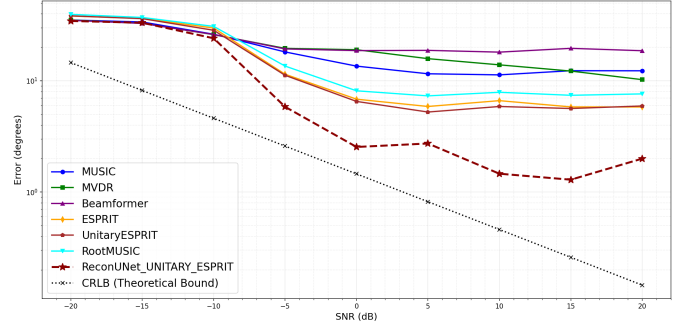


Fig. 8: RMSE vs. SNR (Moderate): 3 direct sources + 1 coherent replica; ESPRIT back end. Thin solid lines: classical on raw SCM. Dashed star-marked line: same back end on reconstructed covariance / predicted EVD.

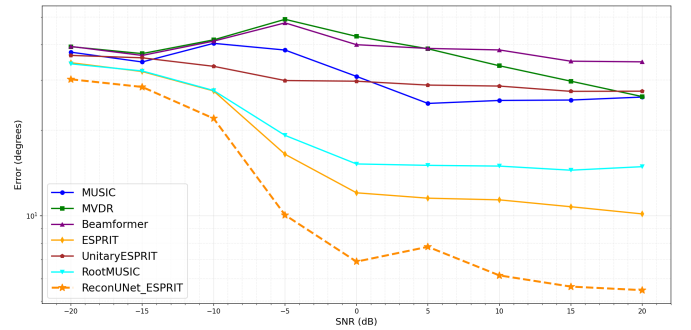


Fig. 9: RMSE vs. SNR (Crowded): 4 direct sources + 3 coherent replicas; Unitary-ESPRIT back end. Thin solid lines: classical on raw SCM. Dashed star-marked line: same back end on reconstructed covariance / predicted EVD.

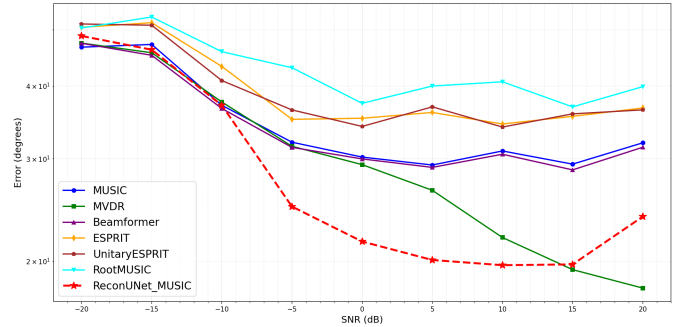


Fig. 10: RMSE vs. SNR (OOD): 1 direct source + 6 coherent replicas; MUSIC back end. Thin solid lines: classical on raw SCM. Dashed star-marked line: same back end on reconstructed covariance / predicted EVD.

MUSIC vs. ReconUNet + MUSIC. The aided curve yields only *modest* improvements in trajectory smoothness and bias; in several ranges the scatter and small under/overshoots persist. Two factors explain this: (i) MUSIC estimates angles through *orthogonality to the noise subspace*, whereas our training explicitly aligns the *signal* subspace and reconstructs a clean covariance—so the noise basis is only an implicit by-product. Small residual errors in the noise subspace can therefore survive and limit gains. (ii) MUSIC still performs a grid scan with

TABLE II: RMSE at 0 dB SNR (deg) across scenarios. Columns: OOD (1+6), Moderate (3+1), Crowded (4+3). Best per column in **bold**.

Method	OOD (1+6)	Mod (3+1)	Crowd (4+3)
Bartlett	29.98	18.68	40.00
MVDR	29.31	19.02	42.78
MUSIC	30.20	13.55	30.93
Root-MUSIC	37.33	8.18	15.21
ESPRIT	35.22	6.83	12.03
Unitary-ESPRIT	34.11	6.52	29.75
ReconUNet + MUSIC	21.62	4.69	14.16
ReconUNet + Unitary-ESPRIT	23.29	2.54	28.29
ReconUNet + ESPRIT	20.89	2.08	6.89

parabolic refinement; even with a better covariance, the sub-grid interpolation error remains a bottleneck. Taken together, these effects make the aided MUSIC less improved than the gridless methods below. This also suggests a path to stronger gains: augment training with a noise-subspace loss or a MUSIC-spectrum consistency term.

Root-MUSIC vs. ReconUNet + Root-MUSIC.: The aided variant tracks the ground-truth lines tightly with visibly reduced variance and far fewer outliers in the sweep (see the zoomed insets). Root-MUSIC depends on stable polynomial roots near the unit circle; when the eigen-gap is small, noise-signal leakage perturbs the root locations and the “root selection” step can fail. The reconstructed covariance enlarges the eigen-gap and reduces leakage, so the correct roots remain dominant and smoothly evolve across angle—eliminating the sporadic slips seen with the classical SCM.

ESPRIT vs. ReconUNet + ESPRIT.: Aided ESPRIT exhibits the largest qualitative improvement: trajectories become nearly collinear with the ground-truth lines and mode-swapping events disappear in the sweep. ESPRIT estimates inter-subarray phase shifts from the *signal* subspace via a rotational-invariance eigenproblem; it is therefore directly helped by our signal-subspace alignment objective and by the Hermitian/diagonal-loading stabilization. The cleaner subspace yields more accurate eigenvalues (phase factors), restoring linear phase progression and producing smooth, bias-free tracks in the inset.

Angles near endfire.: Across all three panels of classical methods, estimates are noticeably weaker as trajectories approach the array’s endfire region ($\theta \approx 30^\circ$ or 150°): the scatter grows, small biases appear, and classical methods occasionally slip. This follows from the ULA manifold $\mathbf{a}(\theta) = [e^{-j\pi(n-1)\cos\theta}]_n$: its angular sensitivity scales with $|\partial\mathbf{a}(\theta)/\partial\theta| \propto |\sin\theta|$. Near endfire, $\sin\theta \rightarrow 0$, so the Fisher information drops, beampatterns broaden, and the signal/noise eigengap shrinks. With our repaired covariance, the endfire penalty is *consistently reduced*: the enlarged eigengap and lower subspace leakage sharpen spectra and stabilize the rotational-invariance/eigen-root steps. In Root-MUSIC and ESPRIT this mitigation is often striking—endfire degradation is faint or, within this sweep, not apparent—whereas MUSIC still shows residual endfire sensitivity due to its reliance on the **noise** subspace and grid refinement. While the front-end improves robustness, the fundamental manifold ill-conditioning near endfire remains a limiting factor.

Summary. Gridless, signal-subspace-based estimators (Root-

TABLE III: Per-source RMSE at 0 dB SNR (degrees) for $K=3$ and $\Delta\theta = 10^\circ$. Best per column in **bold**.

Method	θ_1	θ_2	θ_3
MUSIC	27.66°	6.74°	27.34°
ReconUNet + MUSIC	19.25°	3.71°	15.99°
Root-MUSIC	23.50°	4.88°	23.04°
ReconUNet + Root-MUSIC	5.25°	1.39°	1.20°
ESPRIT	13.61°	2.82°	14.14°
ReconUNet + ESPRIT	1.07°	1.30°	0.89°

MUSIC, ESPRIT) benefit the most from our statistics repair because the network is trained to (i) reconstruct a clean covariance and (ii) *explicitly* align the **signal** subspace. MUSIC, whose decision rule relies on the **noise** subspace and on a discrete grid, shows smaller gains—consistent with the network’s training objectives and the remaining grid interpolation limits.

Quantitatively, these trends are reflected in the per-source RMSEs reported in Table III.

D. Spectrum Comparison at 0 dB SNR

Setup. We visualize (top row) the **MUSIC** pseudospectrum and (bottom row) the **Root-MUSIC** polynomial roots on the unit circle at 0 dB, for two direct sources at fixed angles $\Theta = \{90.6^\circ, 80.1^\circ\}$. Columns correspond to coherent replica counts $\text{mp} \in \{0, 1, 2\}$. The multipath departure angles in each panel are listed as Θ_{mp} .

Findings.

- **MUSIC.** Relative to spectra computed from the raw SCM, the spectrum obtained with our reconstructed covariance exhibits *higher* and *narrower* mainlobes at the true angles, with *deeper* nulls against the noise subspace. Formally, this reflects a larger signal-noise eigengap, which reduces $\|\mathbf{E}_n^H \mathbf{a}(\theta)\|_2$ near the true θ and thus raises $P_{\text{MUSIC}}(\theta) = 1/\|\mathbf{E}_n^H \mathbf{a}(\theta)\|_2^2$. Coherent replicas ($\text{mp}=1,2$) that introduce shoulders or spurious peaks in the raw spectrum are *suppressed* after repair, indicating reduced subspace leakage from correlated paths and improved orthogonality between $\mathbf{a}(\theta)$ and the reconstructed noise basis.
- **Root-MUSIC.** With the repaired covariance, the two DOA-corresponding roots remain *closer to the unit circle* and *stable* across scenarios, while replica-induced roots cluster more interior. This is consistent with the perturbation view: enlarging the eigengap decreases the sensitivity of the signal roots to noise/coherence and stabilizes the root-selection step. In the raw case, coherent replicas can pull the physical roots inward or create ambiguous clusters; after repair, the physical roots dominate and lie closer to the unit circle at arguments matching Θ .

E. Synthesis and Mechanism: Why the Gains Appear

Summary. Across Sections V-A–V-D, a consistent picture emerges: replacing the raw SCM with our reconstructed covariance (or using the predicted EVD) *increases the effective SNR and the signal-noise eigengap*, which in turn sharpens spectra,

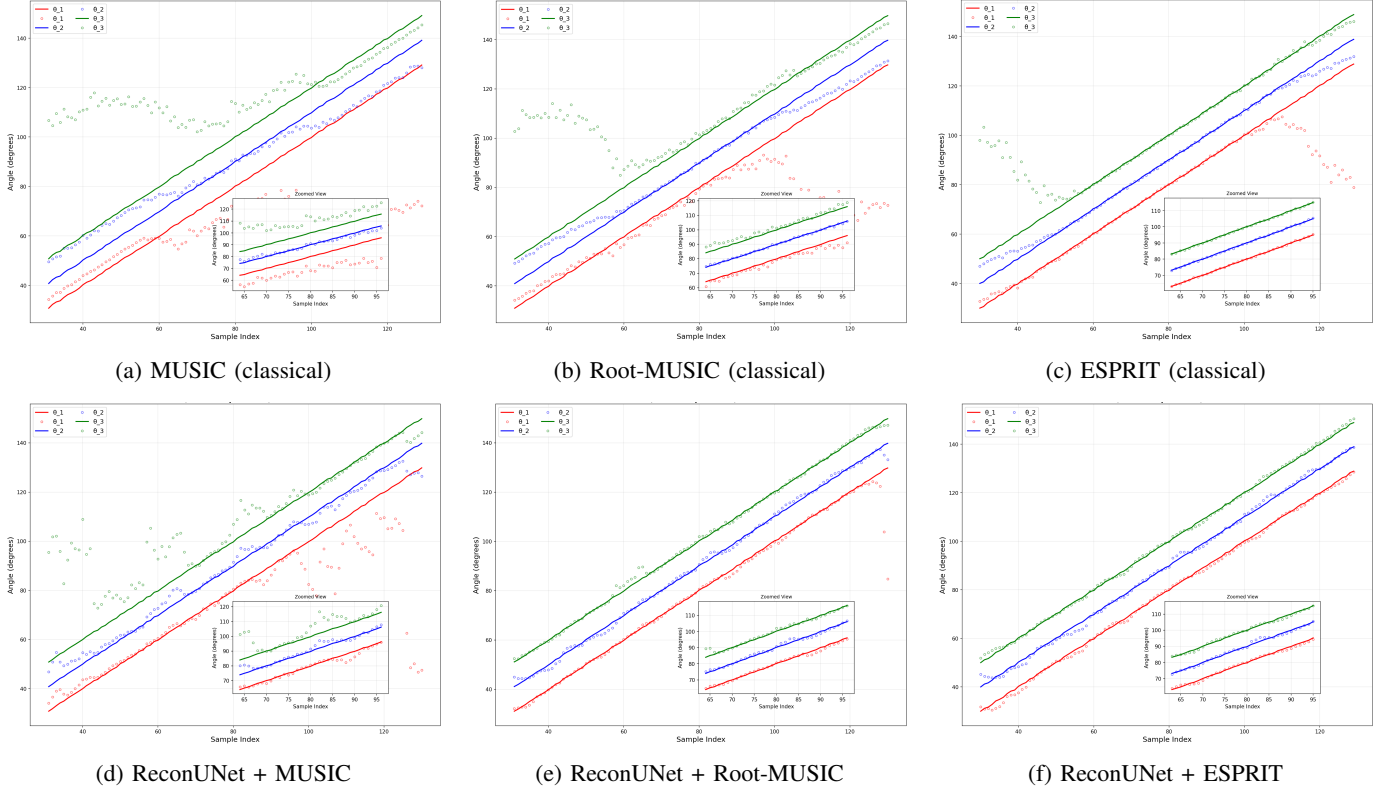


Fig. 11: Angle-sweep at 0dB SNR with $K=3$ and $\Delta\theta = 10^\circ$ (no multipath). Top: classical back ends. Bottom: same back ends using the reconstructed covariance / predicted EVD. Aided versions track the ground-truth trajectories with fewer slips and lower variance, with markedly reduced endfire degradation for Root-MUSIC and ESPRIT.

stabilizes gridless solvers, and reduces RMSE. This behavior is robust across back ends (MUSIC/Root-MUSIC/ESPRIT), scene complexity (from basic to crowded), and even in out-of-distribution (OOD) multipath—with the noted caveat at very high SNR in the OOD case.

Model view. Let the measured covariance be

$$\hat{\mathbf{R}}_x = \underbrace{\mathbf{A}\mathbf{R}_s\mathbf{A}^H}_{\text{ideal target } \mathbf{R}^*} + \sigma_n^2\mathbf{I} + \underbrace{\Delta_{\text{mp}} + \Delta_{\text{imperf}}}_{\text{coherence, mismatches}}$$

where Δ_{mp} captures coherent multipath (rank collapse and cross-terms), Δ_{imperf} encodes array mismatches (gain/phase offsets, position jitter, mutual coupling). The network learns a mapping from the lag-stack features to a *clean, Hermitian, well-conditioned* surrogate

$$\hat{\mathbf{R}}_{\text{EVD}} \approx \mathbf{R}^*,$$

and/or directly to its eigendecomposition. In effect, the corruptions are suppressed (denoised/de-biased), so the top- K signal eigenvalues separate more clearly from the $(N-K)$ noise eigenvalues. A convenient proxy for this improvement is the *effective SNR*

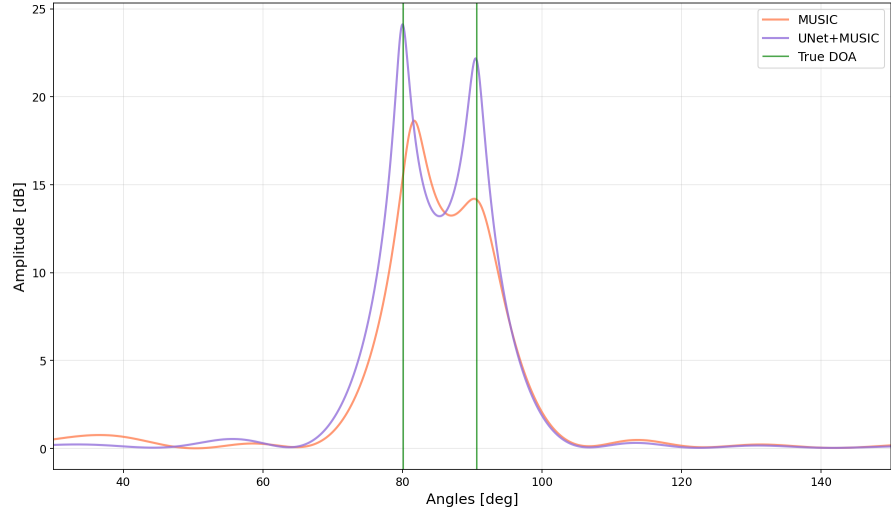
$$\text{SNR}_{\text{eff}} \triangleq \frac{\sum_{i=1}^K \lambda_i(\hat{\mathbf{R}})}{\frac{1}{N-K} \sum_{i=K+1}^N \lambda_i(\hat{\mathbf{R}})},$$

which increases when Δ_{mp} , Δ_{imperf} are attenuated.

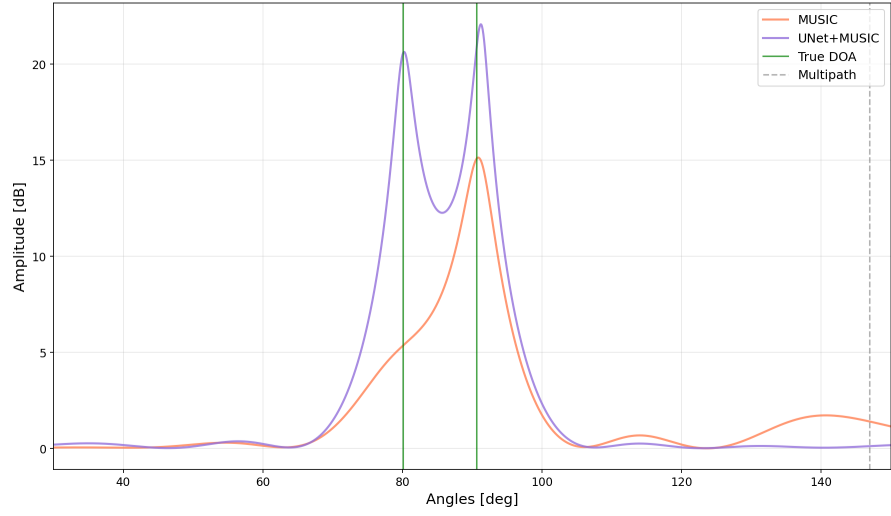
One model for many scenes. Because the network learns to *repair statistics*—rather than regress angles or overfit a

particular back end—it generalizes across the scenario matrix with a single set of weights. No per-scenario fine-tuning is required, and the same front end plugs into distinct subspace engines (MUSIC, Root-MUSIC, ESPRIT), as evidenced by Section V-B.

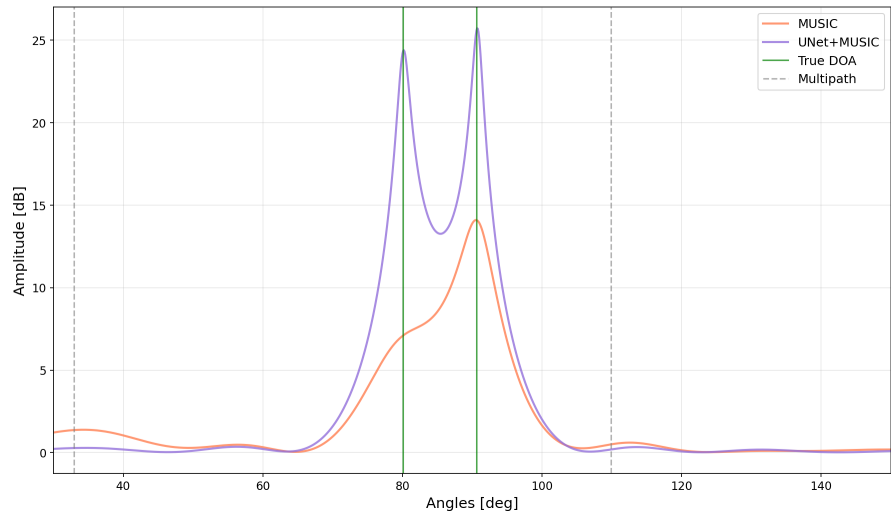
Practical takeaway. Learning to denoise and de-bias the covariance/subspace restores the conditions under which classical DOA estimators excel. The net effect is a higher SNR_{eff} and larger eigengap, which translate into lower RMSE, fewer slips, and improved stability across methods and scenes—while preserving interpretability and the drop-in character required for real systems.



(a) MUSIC, $mp=0$: $\Theta = \{90.6^\circ, 80.1^\circ\}$, $\Theta_{mp} = \{\}$

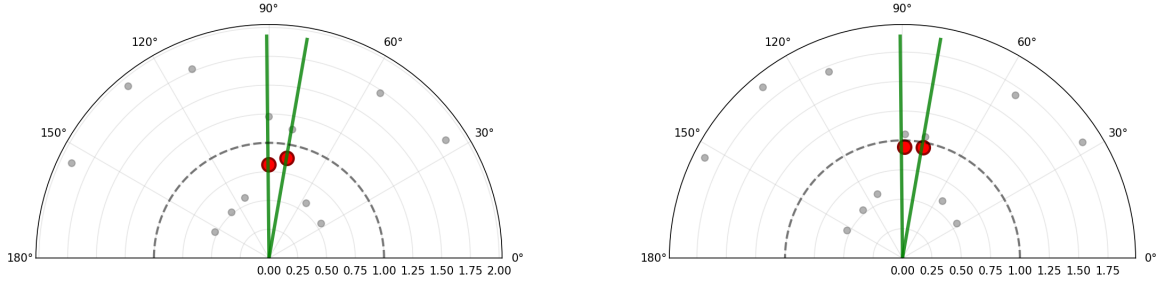


(b) MUSIC, $mp=1$: $\Theta = \{90.6^\circ, 80.1^\circ\}$, $\Theta_{mp} = \{146.9^\circ\}$

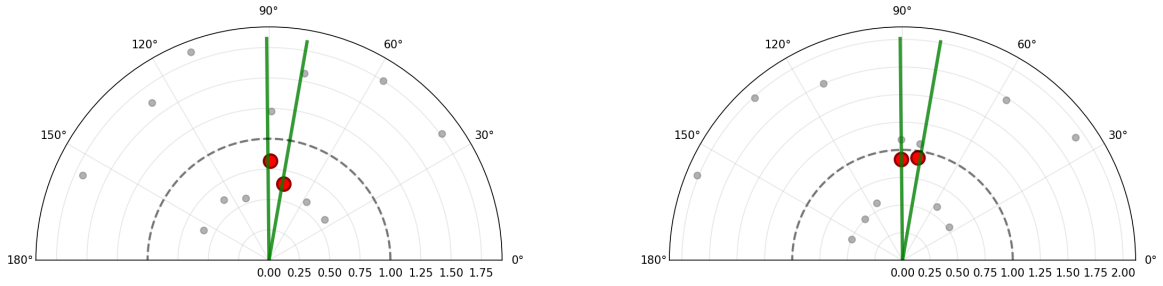


(c) MUSIC, $mp=2$: $\Theta = \{90.6^\circ, 80.1^\circ\}$, $\Theta_{mp} = \{32.9^\circ, 109.9^\circ\}$

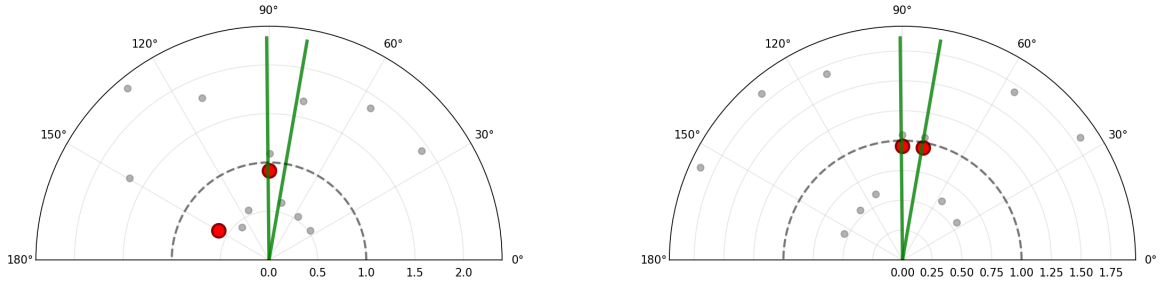
Fig. 12: Spectral views at 0 dB SNR for two sources at $\Theta = \{90.6^\circ, 80.1^\circ\}$ with coherent replicas $mp \in \{0, 1, 2\}$. Panels (a–c): MUSIC pseudospectra.



(d) Root-MUSIC, $mp=0$: $\Theta = \{90.6^\circ, 80.1^\circ\}$, $\Theta_{mp} = \{\}$



(e) Root-MUSIC, $mp=1$: $\Theta = \{90.6^\circ, 80.1^\circ\}$, $\Theta_{mp} = \{126.3^\circ\}$



(f) Root-MUSIC, $mp=2$: $\Theta = \{90.6^\circ, 80.1^\circ\}$, $\Theta_{mp} = \{111.7^\circ, 73.8^\circ\}$

Fig. 12: Spectral views at 0 dB SNR for two sources at $\Theta = \{90.6^\circ, 80.1^\circ\}$ with coherent replicas $mp \in \{0, 1, 2\}$. Panels (d–f): Root-MUSIC polynomial roots on the unit circle.

VI. CONCLUSION

This work introduced a subspace-guided learning front end that reconstructs a clean covariance (and its eigendecomposition) from corrupted, low-SNR snapshots, and then hands those statistics to off-the-shelf DoA estimators without changing their

internals. Across the operating space in Section IV—including coherent multipath and realistic array imperfections—the same trained model consistently tightened error curves and reduced slips for MUSIC, Root-MUSIC, ESPRIT, and Unitary-ESPRIT. The gains are most pronounced where the signal/noise eigengap

is small (low SNR, coherence, imperfect arrays), and they persist under crowded scenes; by design, the approach remains *algorithm-agnostic* and *plug-and-play*.

A central outcome is *generalization across scenarios with a single set of weights*. We deliberately varied the downstream subspace algorithm per scenario (Sec. V-B) and observed similar benefits, demonstrating that the front end learns to repair second-order structure rather than overfitting to a particular spectrum or solver. In other words, we do not require per-scenario or per-algorithm fine-tuning: the same model operates from the basic single-source regime through multi-source coherent mixtures with array mismatch—even when the exact replica counts or angular configurations differ from training.

The results further indicate practical *field suitability*. Our supervision targets are theoretical, closed-form objects (the clean covariance and its EVD on the nominal manifold). These targets exist for any real scene once a nominal geometry is specified; hence, in real data pipelines they can be derived from physics rather than annotation, enabling semi-synthetic or weakly supervised training: render corrupted inputs from measured snapshots, supervise toward the nominal clean statistics, and deploy the repaired outputs within standard subspace engines. This preserves interpretability (explicit EVD), and enables classical confidence diagnostics (e.g., eigengap-based quality checks) without requiring end-to-end angle labels.

Practical considerations.: At inference, the method adds one forward pass over a compact lag-sensor tensor. Users can either (i) feed the reconstructed covariance to any spectrum-scanning back end or (ii) bypass runtime EVD entirely by using the predicted eigenbasis. Because outputs are classical objects (SCM/EVD), integration with existing peak pickers, source enumeration heuristics, and downstream tracking is straightforward.

Limitations and lessons.: Two phenomena bound performance. First, near endfire the ULA manifold is intrinsically ill-conditioned; while the front end mitigates variance by enlarging eigengaps, the geometry still limits ultimate accuracy. Second, in the out-of-distribution high-SNR corner (Sec. V-B, 1+6 replicas), we observed a shallow floor for the aided MUSIC. These suggest (i) explicit endfire-aware augmentation and (ii) training objectives that also stabilize the *noise* subspace or impose spectrum-consistency penalties for grid-based methods.

Outlook.: Further work will extend training to a broader SNR range and to richer combinations of source counts and coherent-multipath replicas, so the model encounters a denser, more challenging operating envelope. We will also investigate generalization across array geometries (e.g., ULA, UCA and nonuniform apertures). Finally, we will validate the approach on real measurement data, comparing against calibration-grade references to characterize failure modes, tune losses, and assess how well the learned statistics transfer from synthetic supervision to field data.

Takeaway. Learning to *repair the statistics*—instead of regressing angles end-to-end—yields a single, reusable model that generalizes across scenarios and slots seamlessly into classical DoA pipelines. The approach improves accuracy where it matters operationally, preserves estimator transparency, and offers a clear path from theoretical targets to real-world use.

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